



Like Guardian Angels: Continuous Location-Sharing and the Production of Safety Among US College Students

Louise Barkhuus^{1,2,*}, John S. Seberger³, Jorge P. Campos⁴ & Anouk Mols⁵

¹The IT University of Copenhagen, Rued Langgaardsvej 7, 2300 Copenhagen, Denmark;

²New York University, 370 Jay Street, Brooklyn, NY 11201, USA;

³Drexel University, 3675 Market St., Philadelphia, PA 19104, USA;

⁴A & J School, 200 U. Richmond Road, SW15 2SH London, UK;

⁵University of Applied Sciences, Utrecht 3584 CH, The Netherlands

Received: 3 July 2025 / Accepted: 26 March 2026 / Published online: 25 April 2026

© The Author(s) 2026

Abstract. Apps like *Life 360* and Apple's *Find My* have made continuous map-based location-sharing common and convenient. Yet, knowledge of how location-sharing is used in day-to-day life remains limited – especially in relation to young adults. Through a combination of interviews with participants and the deployment of daily mini-questionnaires among college-aged users (i. e. 18–22, $n = 15$) in the New York City area, we identify and present three salient aspects of location-sharing practices: (i) orchestration of social activities; (ii) management of social membership; and (iii) production of a “sense” of safety for oneself and others. Our findings point to new social norms emerging in relation to the proliferation of location-sharing apps among New York college students. We discuss such new norms through the lenses of self-disclosure, privacy, and trust-building. Our research scaffolds a more holistic understanding of location-sharing practices among a generation of digital natives and, in return, provides insights into future of designs of location-based technology.

Key Words: Location-sharing, Young adults, Social relationships, Privacy

1 Introduction

I am someone who, until recent events You
shared your secrets with And your location
You forgot to turn it off And so I watch as
you walk Into some bar called The Black
Dog And pierce new holes in my heart You
forgot to turn it off And it hits me I just don't
understand

–Taylor Swift (2024)

In the rapidly developing digital ecology of the 2020s, people now have the option to continuously share map-based locations via apps such as *Life 360* and Apple's *Find My*. Such continuous location-sharing in real-time introduces a score of novel sociotechnical concerns, ranging from problems of ethics and privacy to novel means of social interaction distributed across embodied and digital spatialities. These are concerns where CSCW has had a long running engagement, from discussions and frameworks of space and place (Harrison and Dourish 1996; Harrison and Tatar 2008) and reasoning around the socio-technical gap (Ackerman 2000), to empirical investigations of location sharing (Boesen et al. 2010; Schildt et al. 2016; Jones et al. 2004). But while the notion of location and place has been explored in the past, the updated practices, reasonings and concerns of continuous location sharing has yet to be documented and analyzed from sociotechnical perspectives. It is with this foundation, we engage with the current practices of continuous location-sharing among college students, in the relation to the fragmented discourses of privacy, trust, and safety.

Early experimentation with digital location detection in civic contexts took place from the late 80s/early 90s, however, a subtle change in the sociotechnical milieu of location-sharing began in 2005 with the release of Google Latitude (i.e., the first person-to-person continuous location sharing service). Such significant change continued with the release of Apple's "Find My Friends" – a service originally referred to as "Find Friends" – in 2011, on the now-ubiquitous iPhone. The subsequent twelve years saw a vast proliferation of location-sharing services (Jennings 2023). As such, early technical research on location-sharing has given way to the need to understand location-sharing from a broader and more holistic perspective, now that it is used broadly and in the wild. Researchers – especially those working in the domain of digital privacy – have already begun the production of such a perspective (e.g., Xie et al. 2014; Barkhuus et al. 2003; Toch et al. 2009; Sadeh et al. 2007; Benisch et al. 2011; Knijnenburg et al. 2013; Barkhuus 2012). Yet, the majority of prior work in this area concerns well-bounded (often experimental) systems (e.g., Barkhuus et al. 2008; Mancini et al. 2011; Toch et al. 2010b; Seeburger and Schroeter 2009), or studies where the location sharing was *provided* to the participants rather than the studies of natural use of location-sharing systems "in the wild" (with notable exceptions Boesen et al. 2010). It is this gap in literature that we aim to address, focusing particularly on naturalistic use of continuous location-sharing apps among college-aged students in an urban setting. In this paper we contribute to understanding the everyday use of location-sharing apps by answering two exploratory research questions:

RQ1 How do contemporary young adults think about and engage in person-to-person location-sharing¹ among members of their peer groups (e.g., friends, romantic partners, roommates, etc.)?

RQ2 How do contemporary young adults perceive the notion of safety and security that location sharing among peer groups might effect in an urban social setting?

To answer these questions, we conducted a three-parted study among college-aged location-sharing app users ($n = 15$). Our study involved an initial semi-structured interview with each participant, a two-week deployment of daily mini-questionnaires, and a reflective interview with each participant after they had completed two weeks of mini-questionnaires. Our participant group was chosen for specific reasons: youth, and particularly college students, often lead socially active lives within a broader geographical area, not limited to home and work, compared to what full-time working adults might do (Barkhuus and Dourish 2004). Secondly, members of this population can be expected to encounter situations precarious to safety issues, at a higher rate than adults. Finally, this group of people belong to the so-called generation Z, who have grown up using smart phones as a natural part of their everyday life (McDonald et al. 2024). Our data was analyzed using reflexive thematic analysis, including iterative open coding.

Through analysis and discussion of our findings, we reveal that continuous location-sharing plays a significant role in facilitating participants' daily activities while providing a curious and anticipatory sense of safety for themselves and others. However, such continuous (i.e., "always-on") sharing also introduces challenges related to self-presentation and managing the flow of social relationships – crucial components of young adults' development. In an era characterized by unprecedented technological advancements and evolving sociotechnical conditions that normalize continuous availability, understanding the current landscape of location-sharing practices is essential if we are to understand ongoing transformations in communication, social interactions, and identity expression. Through analysis, presentation of findings, and discussion, we make the following contributions:

- We provide empirical evidence of a sociotechnical milieu in which continuous location-sharing is not only normal, but uncomplicated;
- We emphasize how prior findings about location-sharing practices (e.g., work grounded in the *complexity* of location-sharing practices) require updating in light of the contemporary normality and simplicity of location-sharing; and
- Our discussion underscores the evolution of location-sharing practices and shifting conceptualizations of "safety" among members of Generation Z.

¹ When referring to location-sharing, we mean the continuous map based apps that are most common on smart phones today

We start by situating our paper in related literature, in particular personal directed location-sharing, location inference, and privacy. We then move on to discuss the research aims, how these are addressed, and how the data was collected and analyzed. Finally, we provide the results of the analysis as well as a discussion encompassing self-disclosure, and trust-building dimensions of location-sharing. We provide theory-oriented implications for HCI research about location-sharing, highlighting the need to refresh a priori assumptions in relation to digitally native generations and to consider just what “safety” means in our emergent computational futures.

2 Background and Related Work

Digital location-sharing refers to a heterogeneous set of technology-dependent practices through which individuals may render their location known to others. To ground our research in existing work, particularly as location sharing services have evolved technologically and functionally, we here provide an overview of: (i) directed digital location-sharing (i.e., sharing one’s location with a specific person, group, or set of people); and (ii) location inference based on other digital information. For the purposes of clarity, we break down the former (i.e., directed digital location-sharing) into several categories: broadcast location-sharing, continuous location-sharing, and request-based location-sharing. Following engagement with the literature around directed digital location-sharing and location inference, we give additional attention to a third and broad area of concern related to location-sharing: privacy and people’s sense of safety. Coverage of these areas of research highlights the relevance of RQ1 as presented in Section 1.

2.1 Directed Personal Location-Sharing

Researchers have investigated location-sharing practices since the emergence of early mobile digital systems. Such systems included active badges (Pier 1991) and Intel Research’s *Reno* (Smith et al. 2005). These early prototype systems were generally request-based (Smith et al. 2005; Toch et al. 2010b) or event-triggered (e.g., Bales et al. 2011); there were, however, notable exceptions (e.g., Connecto Barkhuus et al. 2008).

There is a rich history underlying the evolution of smartphone-based location-sharing. While detailed engagement with such a history is beyond the scope of this paper, we provide a few key events to orient the reader.

While mobile phones were more rapidly adopted in Europe and Japan, built-in location technologies were pushed from US based directives, particularly the implementation of “Enhanced 911” in the United States² (Fildes 2026). In 1999,

² 911 is the emergency number in the United States.

the Federal Communications Commission (FCC) mandated what is referred to as “Enhanced 911” (E911) (Federal Communications Commission 1999), which obligated service providers to implement GPS into handsets. This meant that smart phones already had more precise location detection available, as the iPhone spear-headed phones with different applications. Early location sharing system, such as purely broadcasting system, were often part of social media; these included Gowalla, FourSquare and even Facebook where researchers asked questions such as who people shared with Guha and Wicker (2015), with what purpose (Barkhuus and Tashiro 2010) and how they represented their location to others (Rost et al. 2013). Games were developed using location tracking and sharing, sometimes utilizing the “seams” of the location data (Barkhuus et al. 2005; Bell et al. 2006). For an overview of early and mostly now defunct social location-sharing services, see Li and Chen (2010). The next step in the development of location-based application was continuous location sharing, which is the focus of this study.

2.1.1 Continuous Location-Sharing

The ability to continuously share personal location emerged alongside the evolution of smartphones (e.g., improved battery life – initially it was not feasible to have GPS running in the background of mobile phones since it would rapidly drain the battery for other more important functionalities such as phone calls and texting). Most phones today use a mix of GPS, Wi-Fi, Bluetooth and cell phone tower triangulation for location determination, making it generally accurate down to a few meters.

The earliest commercial continuous system was Google Latitude and while it was little known commercially, it was used as an example in several studies (Page et al. 2013; Page and Kobsa 2010, 2009). No known real-world user studies were conducted with Google Latitude, but many studies used their own prototypes. Connecto, for example, was a system that shared the location people had tagged, with selected friends, continuously, except when the user put the system on “manual” (Barkhuus et al. 2008). Other custom-built systems include BuddyTracker which was used for a study of family dynamics, but where the functionality included a notification to people whose location had been looked up (Mancini et al. 2011).³ Note, that already in the case of Reno, which was a request based system, researchers found that this particular functionality sometimes resulted in users *not* wanting to look up another person’s location solely so as not to disturb them with the notification. As such, results of user studies are bound to differ significantly from “true” continuous location-sharing systems, such as the ones common today (e.g. *Find My* and *Life 360*).

³ BuddyTracker, which while described as request-based, was actually only notification-based and as such continuous. But as the location was requested, the system sent a notification to the located user (Jedrzejczyk et al. 2010).

A fairly common way of sharing location is through temporary settings; the messaging app WhatsApp, for example, allows people to share location for a limited time, and Facebook's messenger app is capable of the same. The location is map-based and real-time, within the time limit. One such prototype system was Locacino, a rule-based location-sharing system. Although it was described as being based on requests, the requests were being answered within the set of user-defined rules, and as such functioned as time-limited continuous location-sharing. Interestingly, the users in their study found the rules around time constraint to be less useful than rules based on people (Toch et al. 2010a). As commercial location-sharing tools became more common, researchers also looked at family dynamics (Boesen et al. 2010) and care-taking (Wan et al. 2014) as part of the use of commercial location trackers.

A newer strain of research focuses on location sharing in terms of *one-way tracking approach* (and also specifically names location-sharing *tracking*). A review of 23 papers about "location-tracking", views location tracking as a practice that became common among parents (Davis et al. 2023), with motives ranging from safety, control, worry and a sense of security, to convenience, care, connection, and curiosity. The review does not address how these apps are two-way systems where parents' location is also available to the child and where the child or teenager can use the location sharing to share with their friends as well. Some studies of pre-teens indicate that parents monitor their child's location for assurance (Sukk and Siibak 2021), to guide them Mols et al. (2023), and for convenience in the daily organization of family life (Dungey 2024). In another study only a small sample of parents used location-tracking apps, and tracking teenagers was considered more intrusive and controlling than tracking a younger child's location (Widmer and Albrechtslund 2021). Only a minority of the studies mention how location monitoring of pre-teens often entails reciprocal practices, whereby children also monitor their parents' locations (Dungey 2024; Mols et al. 2023).

Over the past two decades, studies overwhelmingly concluded and suggested that people wanted a wide set of sharing limitations, from time and specific people to the level of location granularity. Prototype developers took for granted the importance of informing the tracked individual that they had been location, this was so important that only one early system to our knowledge did not implement a request or notification: Connecto (Barkhuus et al. 2008). Yet, today, the most commercial systems are exactly what researchers believed would not be acceptable among users: real-time, continuous, no notification, no requests systems, such as Life 360 (Android and iPhone) and Find My (iPhone).

2.2 Privacy and People's Sense of Safety

Most previous research on location-tracking or location-sharing considers aspects of privacy: how privacy should be managed or controlled by not sharing beyond approved friends, how devices and applications should have the option to be turned off and how different people need different location granularity settings.

Early studies looked at preferences, phrased as privacy concerns, for whom to share location with, often coarsely categorized into friends, family and colleagues (Xie et al. 2014; Lederer et al. 2003; Consolvo et al. 2005), or in what context (Barkhuus et al. 2003) or presented prototypes of location-sharing applications as “privacy preserving” sharing applications due to intricate options for setup and self-controlled sharing rules (Toch et al. 2009). Virtually all prototypes, from PeopleFinder (Sadeh et al. 2007), that turned into Locyoution (Tsai et al. 2009), to BuddyTracker (Jedrzejczyk et al. 2010) as well as empirical studies based on hypothetical or expressed practices (Brush et al. 2010; Patil et al. 2015; Knijnenburg et al. 2013), investigated privacy preferences and presented privacy-sensitive design guidelines with the explicit understanding that adoption depended on specific differential preferences.

The broad terminology and common perception of location-based services as a homogeneous notion make it difficult to determine how many people today are using direct location-sharing tools. A recent Harrison Poll, for example, inquired into the use of location-based features but did not distinguish between location-tracking (temporary or continuous) or simply location-sharing features, meaning any application that uses the device's location, such as map placement or social media suggesting a check-in location. So while they found that 79% of the US population uses location features, it was not clear how many of those used location-sharing (Franz and Laningham 2022). Another article discussed how common it is to share location, particularly young people who “tend to be more likely than older adults to adopt new technologies” (Beck 2019). But we do not have any exact numbers on who is sharing location directly.

While most of the research presented here focuses on benevolent sharing, that is, situations where two people agree to share location, or where knowledge of someone's location is through knowing that person well (typically in close-knit families, members already know where the other members are located, through regular practices and mutual informing), location-tracking has also been used maliciously, for example in cases of intimate partner violence situations (Fraser et al. 2010; Bellini et al. 2021) and stalking (Freed et al. 2018). In many cases, technologies can have dual purposes, or have been developed with “good intentions”, for keeping children safe for example, but have been repurposed into spyware and tracking in malicious contexts (Chatterjee et al. 2018). Such re-purposing or simply addition



Figure 1. Diagram depicting the three parts of the study

of functionality, resonates with the notion of “function creep” as described by Lyon (2007).

A core takeaway from past location-sharing systems and research into their use, is that a person’s location, particularly real-time location, has always been considered to be sensitive information by researchers and developers, likely because the digital representation of this enables sharing *beyond* friends, family and significant others, who would already know the individual’s whereabouts. However, instead of approaching location sharing as a privacy-sensitive function, in this study we take a step back and approach our participants with a neutral curiosity, inquiring into their sharing hesitancy as well as their limits of concerns. Our findings contrast many of the early studies, in that the hesitancy is mostly outweighed by a newly found desire for and concept of safety. We will return to several aspects of the work described above in our discussion and consider them in relation to our findings.

3 Methods

For studying the use of location-sharing in-situ, we conducted a qualitative study, based on interviews and self-reporting of location-sharing practices. Our study consisted of three parts: (i) an initial interview; (ii) a two-week deployment of daily mini-questionnaires; and (iii) a second interview during which participants reflected on their use of location-sharing apps during participation in the two-week deployment of the mini-questionnaires. All parts of the study took place between February and early May of 2023 (i.e., during participants’ spring semester).⁴ All procedures received ethical approval from The IT University of Copenhagen, the university of which the first author was affiliated during the data collection. All participants gave consent through a consent form, which was filled in at the beginning of the study; participants were asked again, immediately before each interview, if they would continue to take part of the study, and informed that they were not obligated to continue. At the end of the study, the participants received a gift card of \$25 to their choice of merchant thanking them for their time.

⁴ One of the participants (Charlie) completed one week of mini-questionnaire during his spring break, and another (Anais) was away on a trip with her friends for five of the 14 days, but otherwise, all the participants’ two-week mini-questionnaire took place during regular school weeks, outside exam periods.

We describe the study flow in the following subsection, alongside a graphical representation thereof (see Fig. 1.) After describing the study flow, we proceed by providing detailed descriptions of each part of the study in subsequent sections.

3.1 Study Flow

The initial one-on-one, semi-structured interviews (see Fig. 1, left) introduced participants to the study and helped establish rapport between participants and researchers. The initial interview also helped the research team understand participants' current relationship with technologies and the structure of participants' social circles. In the second part of the study (see Fig. 1, center), participants responded to daily mini-questionnaires for a period of two weeks. We asked participants to engage in this part of the study to promote reflection about how they had been interacting with location-sharing technologies. Upon completion of two weeks' worth of mini-questionnaires, we conducted a second round of one-on-one, semi-structured interviews (see Fig. 1, right) to debrief and reflect on participant use of location-sharing apps. Both interviews focused primarily on interpersonal location-sharing as defined in the introduction. Further, interviews primarily concerned use of *Life 360* and Apple's *Find My* because these were the location-sharing apps commonly used by participants.

3.2 Participants and Place of Study

Our selection criteria for participants was that they were college students, lived in the New York City area and used a location-sharing application with one or more people. New York City was chosen due to its high density of social establishments, and diversity of living arrangements for students. Our goal was to find participants who lived social lives in an urban setting, potentially leading to active use (checking on people they were sharing with) of location-sharing. Students also often find themselves in unfamiliar situations, to a much higher degree than for example a full-time working parents of small children, who lead more regular lives, spending many hours in the same place at a time. Previous research shows that college students often lack a "base" for which they spend significant time (Barkhuus and Dourish 2004), and as such might use location-sharing in more complex ways. They are also less likely to live together with potential romantic partners, which means location sharing can be potentially more useful between romantic partners. We acknowledge that the sample is not a representation of college students in the US, and certainly not students in other parts of the Western world. As such, our inquiries should be understood in this context. Our aim was to investigate how *this* particular set of people utilizes location sharing and how the practices contributes to their specific lifestyle.

Using a combination of purposive and snowball sampling, we successfully recruited 15 participants (initially 17 students participated in the first interview,

but two dropped out during the mini-questionnaire time period), all students living within the boundaries of New York City⁵. All participants were young adult students of New York-based universities (Columbia University, New York University, CUNY (City University of New York) and Pratt Institute of Art). Nine participants self-identified as female and six participants self-identified as male. All participants were between the ages of 18 and 22. At the time of data collection, the average age of participants was 20. Seven of the participants were in a romantic relationship with people outside the cohort studied, but none of them were living together (none of the participants were romantic partners or friends with one another).

Table 1 includes a pseudonymized list of the participants. It also presents relevant demographic information about participants. Participants who were involved in romantic relationships are indicated by asterisks in Table 1. None of the participants had grown up in New York City. All participants had lived in New York City for as long as they had been students there, except Collin, who was a transfer student and had only lived in New York for two of his almost 4 years of studies. All but three participants had grown up in the United States (e.g., the Midwest, South and North East). Two participants were originally from Taiwan and Turkey, respectively, but had spent high school years in the US; one participant grew up in Shanghai, China with a Chinese mother and American father, and attended an international school before moving to the US for university. This means that the only student who could be considered an “international student” in the traditional sense, meaning a student growing up in another country who comes to the US solely to attend university, was Bryan, who had grown up in Turkey to Turkish parents, yet, he has also spent a year in high school in the US. While we did not ask for citizenship details, he would from deduction be the only non-US citizen in our sample.

3.3 Data Collection

In this subsection, we describe the mode by which we collected data during each phase of the study. In sequence, we describe initial interviews, deployment of daily mini-questionnaires for a period of two weeks, and second interviews. As described above, all data were collected between February and May of 2023.

3.3.1 Initial Interview

We initiated the study presented here with semi-structured interviews, conducted on a one-on-one basis (i.e., occurring between one member of the research team and one participant). As is standard in semi-structured interviewing, we asked open-ended questions. The open-ended nature of our questions allowed flexibility in following up on answers (Rubin and Rubin 2011). We developed our interview

⁵ Participants had no relationships either personal or professional with any of the authors or their related research groups.

Table 1. Tabular presentation of information about participants. From left to right, information includes: participant pseudonym, self-reported sex and age, education, the number of people they were sharing location with at the time of the study (extracted from interviews), and the total number of times they looked up others' location during the two-week period between the two interviews. Note that the latter number does not include information about others looking up the location of our participants.

Name**	Sex, Age	Area of Study	No of People	Total Look Ups
Amanda	F, 19	Nursing, 2nd year	5	12
Sadie	F, 20	Psychology, 3rd year	15+	3
Alyssa*	F, 19	Global Studies, 2nd year	7	35
Sandra*	F, 20	Psychology, 3rd year	10	14
Bella*	F, 19	Applied Psychology, 2nd year	5	65
Nathan	M, 20	Public Policy, 3rd year	26	46
Charlie	M, 20	Real Estate, 2nd year	5	24
Cara*	F, 18	Communication Design, 1st	3	19
Anais	F, 20	Photography, 3rd	4	4
Bryan*	M, 22	Film and Media, 3rd year	1	27
Lara	F, 19	Film studies, 1st	1	3
Xavier*	M, 21	Economics, 4th year	3	34
James	M, 20	Political Science, 2nd year	29	40
Collin	M, 22	Engineering, 4th year	9	10
Xiao	F, 21	Computer Science, 4th year	7	28

* The participant had a romantic partner at the time of the study. ** Names have been pseudonimized

protocol based on the recommendations provided by Myers and Newman (2007), focusing particularly on their guidelines such as situating the researcher, minimizing social dissonance and make sure they understood the confidentiality of their disclosures; we also gained insights from pre-study piloting.⁶ Initial interviews (i.e., Part 1 of the study) focused on surfacing insights into how location-sharing apps are involved in the participants' relationships with their friends, roommates, and romantic partners. Participants also discussed location-sharing in relation to their parents and during their teenage years. Results in that relation have been submitted for publication elsewhere.

3.3.2 Daily Mini-Questionnaires

For a period of two weeks after the initial interview, participants responded to a daily mini-questionnaire distributed between 6pm and 8pm in the participants' time zone. Distribution of mini-questionnaires began no more than two days following the completion of the initial interview. In most cases, participants received the first mini-questionnaire either the night of their first interview or the following evening.

⁶ Prior to conducting the study, we conducted a pilot interview with a college student from an affiliate research group. The pilot interview enabled us to fine-tune questions and establish a narrower age range for recruitment.

The mini-questionnaires were distributed to participants via mobile messaging applications chosen by participants (e.g., iMessage, WhatsApp). This information aided the research team in identifying patterns of use among participants. Such patterns of use were underpinned by the emotional and affective information gained through interviews.

The mini-questionnaire was comprised of five questions. Questions probed (1) how many times participants remembered looking at their location-sharing application for the past 24 hours, (2) whose location they looked for specifically, (3) which application was used, (4) whether they were aware of someone else checking their location, and if so (5) who? Responses to daily mini-questionnaire took one of two forms: completion of a Google form or direct reply to the researchers via the messaging app used to circulate the mini-questionnaires. Questions were identical in both cases.

Our method was *inspired* by the experience sampling method (ESM), although it differs in at least one meaningful way; ESM allows “respondents to document their thoughts, feelings, and actions outside the walls of a laboratory and within the context of everyday life” (Christensen et al. 2003, p.53). Yet our deployment of daily mini-questionnaires was less directed at understanding thoughts and feelings than it was at collecting information about the use of location-sharing apps and giving participants an opportunity to engage in reflection.

We limited our collection of questionnaire-based data to one time per day, for two main reasons: Firstly, we reasoned that participants would remember most of their use from that same day. Secondly, we did not want to overburden college-aged participants by pinging them multiple times a day. We predicted that multiple pings per day may interrupt the social lives and practices our study sought to understand, thereby preempting the validity of findings. Given the exploratory nature of our study, as well as the complementary nature of interview and mini-questionnaire data, such a trade-off between fine- and course-grained data is justifiable. The participants were very diligent and generally filled in the questionnaires the evening they received the ping. For two participants, however, 2-3 of their daily questionnaires were filled in the following morning. We contribute the high adherence to the ease of which the questionnaire could be filled in on their mobile phone.

3.3.3 Second Interview

Participant involvement in the two-week deployment of mini-questionnaires was followed by a second interview. The second interview also took a semi-structured format. Yet the structure of the second interviews differed from the structure of the initial interviews: second interviews leveraged data collected through mini-questionnaires to inform the questions and asked about specific situations that the participants had reported.

The second interviews took place one to two days after completion of the fourteenth mini-questionnaire. In two cases, however, scheduling conflicts necessitated a longer gap between completion of the final mini-questionnaire and second inter-

views. In these two cases, second interviews were conducted four and seven days subsequent to the completion of the final mini-questionnaire.

3.4 Data Analysis

During the data collection process, the interview data were examined to identify emerging areas of interest. Although the interview schedule was set, broader questions were asked based on each interview's context. This approach served as a means of confirming and refining construct validity, ensuring that the nuances of each participant were accounted for. By integrating data collection and analysis in this manner, we ensured that each interview's specifics were taken into account without losing sight of the broader context being studied (Klein and Myers 1999).

The audio data were transcribed, and the files imported to a data analysis software (Atlas.ti). Thematic analysis was used to analyse the interview data following Braun and Clarke's (2006) framework. Following an inductive thematic analysis approach (Braun and Clarke 2006), we extracted basic codes that could encompass a variety of practices (such as 'location-sharing for convenience', and 'location-sharing to reduce anxiety'). Accordingly, the analysis involved a primary immersion in the data to gain a better understanding of it. Second, we started generating initial codes by identifying and labelling segments of data with descriptive codes. Third, we grouped codes into potential patterns based on similarities and potential connections. These patterns were reviewed and refined through an iterative process, making it possible to identify subtle themes, thereby enriching and adding complexity to the analysis. Two of the authors conducted the coding, but all authors were involved in the broader discussion of the results.

4 Results

We now take a closer look at the participants' practices of using location-sharing in everyday life. First, we present results from the mini-questionnaire data collection, which provide an overall sense of how often and with whom participants shared location directly and continuously. Second, we turn our attention to data collected through semi-structured interviews. Specifically, we explore three themes that emerged through our thematic analysis of the qualitative data: *(i)* the use of location-sharing to manage social membership; *(ii)* the use of location-sharing to orchestrate social activities; and *(iii)* the use of location-sharing apps as a means of constructing a sense of safety. We present each theme in turn.

4.1 Results from Daily Mini-Questionnaires

The mini-questionnaires revealed that participants checked someone else's location, on average, 1.8 times during the 24 hours preceding the questionnaire prompt.

While a mean of 1.8 may appear to be low, we note that the range of location-checking frequency is wide: An average frequency of 1.8 times per day (i.e., functionally “two”) indicates that checking on the location of friends is, indeed, a part of daily digital life (see: Seberger and Bowker 2021; Edwards 2019) for our participants. A phenomenon need not be frequent at the scale of hours and minutes (i.e., subdivisions of a given day) in order to be of import and interest.) Given the limitations of prior work in relation to the contemporary location-sharing ecology and practices therein (see Section 2), we offer a baseline of roughly two location-checks per day.

In terms of who they looked for, the answers ranged from friends, roommates, parents, and partners, with the most common answer being friends. Similarly, when asked who they thought checked *their* location most during those past 24 hours, the most common for participants not in romantic relationships was a particular friend (sometimes named, sometimes defined, for example “my best friend”). However, the most common daily answer by participants who *were* in a romantic relationship at the time of the study, was that they looked up their partner’s location and that they thought their partner looked at their location. There were two exceptions to this rules: Bella and Sandra, who were both partnered, looked more times for their friends’ locations than their boyfriends’ and Bella and Cara reported only few instances (4 and 3 respectively) where they knew their partner, or anyone else, had looked up their location.

When asking who the participants thought or knew was checking their location over the time period, participants seemed uncertain how often this actually happened and tended to err on the conservative side. Since we did not want to put extra effort on the participants (and ethically could not include their friends in our study by asking our participants to ask them), we had phrased the question broadly: “Did you notice if anyone checked your location during the last 24 hours? If yes, who?” The most common answer by far was “Not that I’m aware of”, yet occasionally they knew for certain that a friend, parent or partner had checked on them because they had communicated about it. Charlie for example wrote “I did notice that my friend [Lindsey] tracked me (she told me about it)”. So essentially our participants reported only between 0 and 1.6 people on average per day, a median of 0.5, who they knew or presumed had checked their location. The delta between reported known incidents of *being* checked-in on (median = 0.5, range = 0 - 1.6) and the reported frequency of checking in on others (median = x, range = y - z) highlights a basic uncertainty surrounding the use of location-sharing technologies among members of a given social circle. Generally, participants with partners checked on their partner’s location to a higher degree than friends, and the participants without, checked mostly on other friends’ location. All participants used Apple’s Find My, a few also used Life 360, and nobody shared location on SnapChat, but a few had previously done that. One person occasionally shared her location in a time-limited way on WhatsApp.

4.2 Location-Sharing as a Way to Maintain Social Connection

We now report results from our analysis of the interviews starting by looking at how social relationships are maintained through location-sharing and how location-sharing reflects strong social bonds. Our focus is mainly on people's everyday social circles including friends and partners, rather than sharing between them and their parents, which around half of the participants did.

For some of the participants, location-sharing was seen as a tool representing their belonging to a group. Being part of a group of people sharing their location was perceived as a form of social belonging and cohesion. We explore first how the participants sometimes were very aware of this, and second, how other times, the sense of belonging emerged more subtly.

Charlie for example told us how location-sharing was used differently with different friends. Where some, perhaps closer friends 'checked in' on him, and vice versa, other friends encouraged him to use it more actively when coordinating socially. On the other hand, Nathan prefaced by saying he "feel[s] like [he is] not the norm on this", he described a normal day as him starting out by checking his friends' locations:

"So I have a tendency to [...] check locations, [...] see where everyone is, or if, if someone is like super close to me, like on my block or something, I'll text them asking if they want to go out and get like coffee or breakfast and then do something accordingly."

For others, like Sandra, the sense of social cohesion through location tracking was more narrow than her broader social group, instead, it was the connection with her best friend, with whom she was close and hung out with almost every day. She explained:

"I do tend to track my friend [Kylie] a lot. [...] I guess it's out of convenience, [...] Like if she's not answering, like, are you in class? Are you sleeping or are you, like, doing something? Are you hanging out with somebody and not [in] a creepy way or like a way that like, I'll get mad if you're, doing nothing and you don't answer me. I'm just like, more of a out of curiosity. Like, why aren't you answering me? [...] am I calling you while you're in class right now kind of thing?"

Finally, for Sadie, location-sharing was initially a tool for her and her roommates⁷ to be able to just check in on each other for safety, but they kept the Life 360 circle after she moved to live on her own, but also said that location-sharing was still useful for her social interactions with the former roommates:

⁷ Note that in the US 'roommates' is a term that is used broadly, to mean both the people a person is sharing an apartment/house with, but also a person they are sharing bedroom with, which is still a fairly common setup in college dormitories, particularly in high-cost living areas. We mention specifically if the participants made us aware of which type of roommate that is referred to further down.

“I just moved around the corner and one of my former roommates is in London right now for study abroad, but the other roommate is back at the apartment. So we, like, see each other at least once a week. We get coffee and we hang out and stuff like that, because I do like miss seeing people. But just, I think especially living in New York, like being surrounded by people 24/7 was just too overwhelming.”

This quote illustrates well how a set of social relations where location-sharing starts out as practicalities and a motivation of safety, continues to provide social belonging, for people moving far away, as well as relations staying close by.

We now continue looking closer at a more practical use of location-sharing, something that has often been highlighted as a key feature in relation to these services.

4.3 Location-Sharing for Managing Daily Life

One of the most mundane uses of location-sharing is managing daily communication and activities; it has been illustrated in the past, how people enjoy how location-sharing can support micro-coordination and saves on more direct communication (Barkhuus et al. 2008; Schildt et al. 2016; Bales et al. 2011; Mancini et al. 2011). In extension of these past findings, we sought to investigate how our college students were using it to manage their daily lives. Several participants discussed using location-sharing apps to decide when they should leave home, when to expect a friend's arrival, and when to call others, amongst a plethora of other daily organizational tasks.

College students often lead nomadic but at the same time also *regular* lives (Barkhuus and Dourish 2004), leading them to need to coordinate meticulously with friends for socializing. Some participants had very regular activities where they utilized location-sharing, like Nathan who explained his Tuesday and Thursday mornings:

“I have a class on Tuesdays and Thursdays with six of my friends and we all kind of come from [...] the same triangulated location of, like, eighth and Broadway and St Mark's and like Union Square. So what we'll all do is we'll try and, like, meet in the middle and then, [...] with the class starting at 11, we'll all be there at like 10:50. But in reality that never happens. So you kind of pick two people, track their location, see when you all get there, and then go [...]. Or [...] I'll check their location, shoot them a text. and then go: Hey, can you grab me a seat and I'll meet you there.”

Cara talked about how it had worked with her and her three roommates the previous semester before she moved to a single suite this current semester:

“We all had each other shared and we were using it for the purpose of almost, not *getting rid* of communication, but just a quicker, more, I guess, technologically advanced way of just communicating. [Like] you can just see that they are there,

or their class I think ends at 6:30 so they're not gonna be back yet. Oh are they already at dinner, or should I grab something for them?"

Yet, the living situation did not work out for Cara who moved to a single dormitory room and stopped sharing location with her previous roommates (as opposed to Sadie, Cara did not get along with them in the long run).

Several of the participants found examples of ways in which smaller tasks were supported, but also *ad hoc* socializing through checking local friends' locations. Sadie provided an amusing example of how location-sharing could help her control her shopping habits:

"[S]ometimes if I'm home⁸ or like I have to go run errands, I'll check it to see if, like my friends who live near me are also home and can go run errands with me. I'm like, I don't want to go to Target by myself because then I'll just buy way too many things at target. Like, I need someone to tell me to stop."

The participants with partners also used it substantively for coordinating and getting information on when they should depart for a meet-up, or when they would expect their partner to come over (none of the participants lived with their partners, a setup corresponding well to their demographics as college students). Sandra explained her most common use of the app as tracking her boyfriend:

"I like using the tracker when my boyfriend is on the way to me so I can kind of plan out when I'm gonna go home. Like if I'm at school studying and he's on the way, it takes like [...] 50 minutes to an hour for him to get here, depending on like the traffic and the weather and how fast you drive. So I like to check it when he's on the way so I can know, [...] like when to walk home so we [...] get to my place at the same time."

Some of the participants were sharing their bedroom with a roommate, and if the roommate was in a romantic relationship, it was not uncommon for the participants to indicate that they used location-sharing to find out if their roommate was home, or if they were far enough away that they were 'safe'. Bella for example, who shared location mostly with her boyfriend and roommate (who she was sharing a bedroom with), alluded to how she was using her location tracker by describing how she not only tracked her roommate for when she was close to their dorm, but also knew her class schedule for planning and knowing if she would be back shortly. This was essential for her knowing how much time she would have to herself "or whatever", alluding to spending intimate time with her boyfriend.

Xavier was being even more explicit when talking about how he mainly used location-sharing with his roommate to find out if he was home. Early on in the interview, he explains his situation:

⁸ In this instance, she is referring to home as her parents' house in Connecticut.

“I live in a suite with [...] in a five-person suite comprised of three singles and two doubles [sic]. And I live in the double. [light chuckle] Okay. Yeah. Which is. You know, why it’s. It’s [...] It’s especially useful to have my roommate’s location. And then. Yeah, we also have, like, a common space.”

Later, he continues down the same thread:

“So, I mean, it’s just nice to know whether he’s in the room or not. And specifically [laugh...] we are both in pretty serious relationships. In his case, his girlfriend lives off-campus and has an apartment. And he spends most nights over there [...] So, you know, normally I assume that he’s not going to be in the room. [...] There, there’s the case when I’m planning on having my girlfriend in there. And then it’s like, okay, this is sort of [I’m wondering] whether my roommate is home. Normally he is not, but if he is, then you know, I’m not going to have her over [...] So it’s mostly that. It’s definitely, not [...] that I’m] looking to see where he is on campus and going to follow him. Just to know whether the room is vacant.”

In these examples, sharing was accepted as a way of managing private space and personal (romantic) relationships.

A significant contrast was illustrated by James, who found that while there were social benefits to location-sharing, it could also be used for avoiding socially uncomfortable situations. He told us how some people he had presumed were his friends had started ignoring him, and that he fell out with his roommate during the semester. When he was interviewed, he actively avoided talking to his roommate and the roommate’s friends; the location-sharing app was useful for that:

“Well, like, sometimes I’ll [check] is my roommate in the room? Like, is it... especially since we’re not really that close any more. Like, is it okay to go back and like change real quick or grab stuff without having to interact with him?”

James also explained how he would use his location-sharing for meeting up with friends (“Like, how far, like how long it’s going to take for them to get there, how far away they are.”), but when asked about whose location he checked the most, it was his roommates by far, and vice versa. These examples illustrated how the location-sharing tools were woven into everyday activities for negotiating everyday activities and micro-coordination. James’s use of location-sharing technologies compliments prior findings (see Section 2 about the extent to which location-related information can be used to infer non-location-related aspects of a person whose location is tracked).

4.4 Location-Sharing as a Way of Ensuring a “Sense” of Safety

When inquiring into motivations for participants to share location with their friends, roommates and acquaintances, all participants cited safety as a primary reason. While most (12 out of the 15) had already used location-sharing in high school or even middle school, either with their parents or friends, the number of people

they shared it with had increased as they moved to New York City for university. Sometimes a roommate had suggested sharing, or they had suggested it to their friends themselves, as they had arrived for university. Sadie provided a typical explanation:

“And then coming to college, [...] I started sharing my locations with my roommates. And then we made a really good friend who became our roommate sophomore year, and we started sharing our locations with her. And that was more for safety reasons, just like because we’re in New York. [...] like, if anyone’s, like out walking late at night by themselves or something [...] you could just send a text to the group chat, like, hey, just a heads up, I’m like, by myself. If you could just check my location occasionally, [to see] that I get home safely and something like that.”

Overall, the notion of safety was an overarching theme when talking about reasons for sharing location, but also when mentioning specific situations, all 15 participants mentioned at one point or another the issue of safety. To analyze this topic in detail we explore two sides of it: the participants’ own safety and their perception and concern for others’ safety. We present results from each side in turn.

Our encounter with safety as a prominent facet of location-sharing among the college students we interviewed presents as a tangle of issues generally not considered as a set: privacy, trust, and safety. (See Section 2.) To undo such a tangle, we devote the next two sections to understanding participant perceptions of individual safety and the safety of others. This analysis serves as the foundation for a deeper discussion of continuous location-sharing in Sections 5 and 6. We choose to delay the provision of deep discussion of findings to such sections in service to readability.

4.4.1 *Safety of Oneself*

It was clear that the location-sharing was described as a safety tool for the participants themselves (again, all 15 participants mentioned safety during the interviews). They phrased it as useful in potentially extreme situations where they would go missing or not come home. Bella, for example, explained:

“Yeah, I think I first started sharing my location when I came to NYU, and it was like, when I used to go out on, like, dates or stuff like that, just knowing that, [...] many people had my location [...], I’d have like maybe three friends, and I’d be like, Hey, I’m going to go out, [...] if I don’t come back or if I don’t check in, you know, you know, where I am, Please come find me. That’s always like been helpful.”

Xiao reported using a similar example for when she and her friends went on dates.

Cara acknowledged that she lived in a fairly rough part of the city⁹, which was where her university was located. When we interviewed her, she had moved out of university housing and into a private building, where the university shuttle did not run close by, but she could still take it part of the way. She explained how she felt the location-sharing helped her maintain a sense of safety, despite not directly communicating with her friends about it:

“[L]ike the other night, I, it was like four in the morning. It was during midterms and I wasn’t getting back to my apartment from on-campus, technically, from my friend’s [apartment], from working on a project until like four in the morning. And I took the shuttle, so that was good. But like, she has my location and I definitely I know that, like in the back of my mind [...] she knows when I’m getting back, she’ll probably check it.”

Sometimes participants described situations in which they were uncomfortable in terms of safety but where somehow, location-sharing helped mediate the sense of safety in the situation. Nathan, who had a rich social life, explained how he had ended up in a frightening situation around four months previously (note that we have elected to include the full direct quote to let Nathan himself describe the situation in detail):

“I went on a date during finals season last year, if we can call it a date. It was like two o’clock in the morning. And the person who lived on like 66 [66th street] and said that they were a law school student, and I just got, like, a weird feeling. And I’d gotten there and was like, entirely not what I expected. They were like, hopped up on drugs, like [...] crazy stuff. But I had texted my roommate, [...] that I knew were up because it was finals [...] And then the thing that ended up getting me out of like the bad situation was I had made them aware that my roommate was waiting for me to get home and that my roommate had walked in the door, let’s say like 4:45. And she texted me like, I’m home. Come home now. [...] And I was able to show my phone and be like, Oh, I’m so sorry. Like, look, my roommate’s at our apartment, like, you call the Uber here. I’m like, My roommate’s waiting for me and like, I can’t be here any longer. [...] and I ran out of the door and hopped into the Uber home. [...] Having people’s location in the back of my mind definitely made me feel a lot safer [...] because they [the date] were like, you’re not leaving. Like you’re going to stay with me. We’re going to go to, like Jersey tomorrow morning, Like, I want to take you

⁹ New York City, particularly Manhattan, where all except the students at Pratt lived, is generally very safe, but has, particularly post-pandemic, pockets of ‘rough areas’ where transients and sometimes people of unstable mental health congregate. The area of FIT has recently become such area, despite traditionally, being part of the affluent area Chelsea, it was viewed as ‘nice’. At the same time, FIT is merely blocks away from the newly developed Hudson Yards where new buildings of 20+ floors are sprawling, catering to high income people, rent often being \$10,000+/month. Cara moved to an apartment in one of the private new buildings, after moving out from the place with roommates.

to my little cabin in the woods. And I was like, Oh my God, this is how I die, [...] it was so scary.”

While it is hard to discern to which degree Nathan was in real danger, his knowledge that his friends knew where he was, was the key to feeling safer. We thus encounter a direct linkage between discourses of privacy (i.e., narrowly construed as control over who has access to one’s information) and the location-sharer’s sense of safety.

Finally, Charlie who had a close-knit social circle that included four other friends, described situations where he wanted to go out, but his friends didn’t. Instead, Charlie described his friends as watching over him like “guardian angels” via their location-sharing apps:

“I feel it was good to let them know that I’m going [out], [...] like, that’s where it would be [...] it’s almost like they’re a guardian angel watching over, like making sure nothing bad happens.”

In the light of these quotes from Nathan and Charlie above, we begin to develop a detailed account of location-sharing practices among our participants.

We now take a closer look at location-sharing as a way of making sure others are safe, and what activities the participants performed to ensure others’ safety.

4.4.2 *Safety of Others*

Many of the participants were encouraged to use location-sharing with friends, partners and parents for their own safety, meaning as a way of mitigating others’ concern for them. Of the fifteen participants, eleven mentioned such issue; the four who did not, were all female participants. While sometimes they expressed how that invoked a sense of safety as described above, in other situations, the use was solely accepted as a way of satisfying another set of people’s worries about them without actually improving an individual’s safety. Amanda, for example, who had parents that lived in the Midwest (a much more rural area of the United States), would occasionally hear from her worried parents who she was sharing location with:

“[Y]esterday I went to Queens. [...] And it was maybe around 5 p.m. And my parents called me, all worried because I’m in Queens and they are just, they’re not controlling, they just get worried ¹⁰. And, you know, I understand. [...] I was an Uber and it [the locating sharing app] notified them that I was in Queens. And so, you know, because they didn’t want me to be out too late and everything like that, [...] I know they’re just trying to look out and stuff, so I don’t mind that they have my location as long as it makes them feel better. [...] It doesn’t restrict me from, like going to places I want to go to.”

¹⁰ Queens has often been presented in public media as a troubled and crime-filled neighborhood, which the participant is implying; yet, queens does in fact have a lower crime level than Manhattan where the participant lives (NY State, Division of Criminal Justice Services [2026](#)).

Amanda had accepted that her parents (who she pointed out, had not grown up in the US, and who were not familiar with the different areas of New York City apart from popular culture) liked to be ‘following’ her closely as to stay in the know about her being safe¹¹. While Queens in reality is a diverse Xavier found himself in a similar situation with his girlfriend, who he knew was checking his location significantly more than he did with hers, to make sure he was safe. He explained that she used it much more than him, with probably 20 people on her sharing list, explaining: “and part of that is, is just like safety. I mean it’s, it’s not safe to travel in New York City as a woman.” We will provide a deeper consideration of this construction of “safety” in Section 5.

Another part of *safety for others* was the flip side of people’s own sense of safety that location-sharing provided: participants offered numerous examples of situations where they felt they could help protect someone else or make sure they were safe, by checking in on their location. Collin, for example, explained how he had gone out with a group of friends on someone’s birthday:

“it was just getting [...] pretty late in the night. So some people wanted to head back. Some people were still wanting to stay. [...] I was kind of just checking in and making sure my roommate and some friends got back to their [...] apartments, and dorm, safely. And kinda just making sure everyone got back.”

Other people had more dramatic examples of them being really worried:

“Like my friend Michelle, her phone said that she was in Queens in like the middle of the night, and we all thought that she was like, dead, she wasn’t answering. But I mean, [...] she left her phone in the Uber, and the Uber went home to Queens, and that was everything. Yeah, everything was okay. But that was a really big, like, safety thing because at least we could have seen, like, where she was, had she still been in a random person’s Uber or something. So I think that it really is just like a safety factor.”

Notably, every participant talked about location-sharing as a safety feature, most of them arguing that this was the main reason they were sharing location and using this type of application. They highlighted safety for themselves, their partners, their friends and in one case a participant’s (Xiao) mom who came to visit her in New York from abroad. They ensured us that they were sharing only with trusted people in their lives, except in one case: James said he wanted to delete many of his 29 people but was worried about the potential social backlash it would invoke, when they found out he deleted them.

Participants were generally adamant that location-sharing had increased their safety. However, we found no examples of that specifically. Even in the situation

¹¹ There is a lot more to unpack in terms of parents and their sense of control through location-sharing, however, we are limiting that discussion in this paper, since it is not the focus.

that Nathan had ended up in, described above, it is highly unlikely that his friends would have made it in time if his date indeed had had malicious intent. So the essential factor is that location-sharing increased participants' *sense of* safety and gave them a heightened sense of security. While the majority expressed how they were completely fine about the sharing (they had accepted it themselves in the first place), the cost of using these tools was still the sense of being observed, which some of them discovered and were able to articulate in detail. That brings us to the last topic of our findings, how participants negotiated and maintained their own and others' sense of privacy.

4.5 Respecting and Maintaining Privacy

One important bias when analyzing how participants felt about sharing their location continuously, is of course that they were asked to not only talk about it but also write down how much they looked up others' location for a couple of weeks. During the final interviews, we always asked as one of the first questions, if they felt their habits had changed over those weeks, and while all participants said they did not think so, it is impossible to disregard some potential bias. A few participants actually did report noticing how much their friend or partner looked them up: Bryan for example said that he was surprised to find out just how many times his partner looked up his location since he asked him explicitly to be able to answer our mini-questionnaire.

Broadly, participants were very aware of who they were sharing location with and Xiao for example said that she had never had anything but positive experiences with it. Not even "just neutral experiences", but because she only shared with people she trusted completely, she exclaimed that all experiences were positive. Still, a few participants described instances where people in their sharing list overstepped lines of inquiry or 'nosiness'. Nathan for example found it intruding that his mom was commenting on his whereabouts when he was in California for a few days to visit family. He explained how his mother had commented on his specific whereabouts:

"I knew, like [her] constant checking of the location every time I would move or walk or refresh: 'why are you walking? Why are you Ubering, what's going on?' And that definitely, like, made me feel uncomfortable about the experience."

Interestingly, we found no situations where our participants felt intruded upon by friends or acquaintances on their list, possibly because of the lack of control structure between them (as opposed to a parental relationship, which features a built-in hierarchy) and the fact that they had accepted sharing with friends in the first place, and knew, at least theoretically, that they could delete them at any point. The only example we found of someone "hiding" their location during their two-week period was Alyssa who went to get a tattoo and hid her location by turning off sharing temporarily so that her mom would not find out.

In relation to maintaining privacy, some participants explained how they had needed a warm-up period when initially starting to share their locations continu-

ously. Bryan, for example, had only started sharing location two years previously when his partner, three months into their relationship, asked him to start using it. At first, he felt a bit strange about it, but after a few months, he felt it was okay. At the time of the interview, he said that he did not mind at all, but most of the sharing was with his partner who checked on him, not vice versa.

Others were surprised to find out how much their friends were checking up on their or others' location, compared to themselves. Charlie explained how his friend Lindsey had told him how much she checked the app:

Charlie: "[She] told me, she was like, 'Oh, whenever I'm bored, and I'm just at home, I check to see if you're at home and just, like, chilling or doing homework or whatever. And I message you, but if you're busy, I'm not going to, like, reach out to you, because you're doing your own thing.' And I'm like, Okay, that makes sense."

Interviewer: "Did you know that before she did that or was this new to you?"

Charlie: "No, I had no [idea]. But then actually, before this whole [...] experience [...] one of my friends told me that one of her friends would like, they would share their location with each other and, um, like out of the blue, she would show up to a place where she was at, at that moment [...] And I was like, No, I totally get that. That's like boundary issues. Yeah. So like, in some ways, people do it in different ways. [...] That's kind of creepy though, to think about."

When discussing the issue of privacy and participants' feelings around being 'tracked', it was clear that they mostly thought about it as a trade-off. With a core motivation of utilization being safety, they accepted that it occasionally felt intrusive to have people, no matter how close they were, knowing where they were 24/7. Collin expressed it distinctly:

"Location-sharing is good and important for the safety aspect if anything happens. Like, somebody knows where you are. I've had certain instances that gets a little invasive and somebody would say, What are you doing, What are you doing studying in this place? I'm just kind of minding my business. I feel a little weird [...] But... It's kind of, in a sense, like a necessary evil; I would say that it's evil... but it's a good precautionary tool."

In essence he acknowledged that it is "evil" to have access to such detailed information about others' whereabouts, but because it helps keep them safe, the evil is okay. Yet Collin's use of the phrase, "precautionary tool," is telling: in each of the situations described by participants in which they referenced the improved sense of safety that using location-sharing apps gave them, it is unclear that distributed knowledge (i.e., among a person's circle of friends) of a person's location would have actually prevented harm (i.e., maintained the safety of the user).

One specific situation that came up in our conversations where a participant felt "odd" about being tracked did not involve explicit location-sharing. Cara was

a dog-walker in her spare time and when she started walking a particular dog for a customer, she started getting notifications on her phone that an AirTag was following her. She quickly figured out that it was the dog who was wearing the tag, but she was a bit surprised that the owner had not told her. When she asked, the owner confirmed that the dog had an AirTag in the collar and Cara explained that she understood the owner's reasoning behind having that, but she would still have liked to be informed. She acknowledged that the dog owner could track her anyway¹², but it was strange to get the notifications without being prepared.

In summary, the participants used two arguments to explain why they did not feel the location-sharing was privacy-invasive or intrusive: firstly, they had chosen to share with specific people themselves, and they argued that they trusted every single person on their list. One exception was James, who still shared with people he said he would like to take off, but he felt it would lead to questions and animosity if he deleted them. The second argument was that if any privacy intrusion exists in relation to location-sharing, such intrusion is justifiable because the use of location-sharing was perceived (whether realistically or not) to improve individual safety, their own and others.

In the next section, we discuss the results presented above. In particular, we consider the construction of feelings of safety (i.e., an experiential *sense* of safety) in relation to work described in Section 2. Subsequent to discussion, we provide implications for research in the CSCW and CHI community.

5 Discussion

In our discussion, we take a closer look at how young adults manage to negotiate self-disclosure and privacy concerns on a daily basis, in a sociotechnical world where digital personal information is constantly available. We note that participants' implication of their safety in relation to location-sharing complicates historical findings about the sensitivity of location information (cf (Guha and Wicker 2015)).

We unpack our findings in three inter-related ways. First, we consider the role of self-disclosure and privacy concerns in relation to how participants used and perceived of location-tracking apps. Second, we describe participants' use of location-tracking technologies to *build* trust (as opposed to enrolling trusted friends into location-sharing). Third, we explain the paradoxical use of location-tracking technologies to produce "safety," even though such a condition of "safety" as described by participants was hypothetical.

¹² Cara used a dog-walking matching app, similar to ride sharing, where the app had permission to share her location with the dog owner, as prevention of theft and to provide a greater sense of accountability.

5.1 Self-Disclosure and Privacy Concerns

While it seemingly has become more common to share location with other people, with a self-defined promise to increase a sense of safety, this also raises pertinent questions about self-disclosure in relation to privacy concerns. Prior work (Guha and Wicker 2015) highlighted the perceived privacy-sensitivity of location-related data. Our participants managed to negotiate self-disclosure on an ongoing basis, despite not putting too much thought into how others were looking up their location – such a finding is internally contradictory according to prior work (Guha and Wicker 2015), yet it appears sensible when one considers that the population to which our sample belongs (i.e., college students in large urban settings) inherited a sociotechnical milieu in which location-sharing capabilities have crept (Lyon 2007) into pervasiveness by way of smartphones.

Participants mostly had little sense of who or how much someone else had followed them (digitally) on particular days. Such a claim is supported by the delta between their own location-checking behaviors and the reported behaviors of other participants. A general sense that their location *could* be checked was prevalent – and expected given participants' extant use of location-sharing apps prior to their involvement in the study. Yet the delta between actual behavior and expected behavior presents as a promising avenue for future research.

Other social dynamics emerged when participants explicitly shared location with someone else, as in the case of digital logistics when organizing a meet-up. We heard about how someone was trying to find their mom at the airport, how someone needed to find their partner at a large outdoor event, and how someone was meeting up at a place where construction suddenly made it impossible to take the usual route, and location-sharing 'informed' the other person of the delay. In those situations, self-disclosure was expressed very literally and voluntarily, and the perception of privacy was only present in that this information was shared solely with the selected person (the information was shared at the same time with everybody else on the person's sharing list, but from the person's own perspective, this was the active self-disclosure point). Such privacy-related findings mirror findings from related work (McDonald et al. 2024) in which teen smartphone users were less concerned about general surveillance (e.g., from the government or corporations) than with privacy invasions stemming from people they know or simply other users.

The participants generally expressed feelings of being well in control of their self-disclosure of location and whereabouts, yet half of them shared location with one or more 'old friends': friends who were either not local, or friends they did not see anymore, mostly old high school friends that they had lost touch with. Such a finding constitutes a direct furtherance of prior research about location-sharing, as prior work does not consider how location-sharing practices evolve in time alongside the dynamics of users' social circles. Given the rapidity with which location-sharing has become a pervasive function in worlds mediated by smartphones, it stands to reason that prior work has not considered the social

evolution of location-sharing practices. The historicity of our work (i.e., the point in time at which it was conducted) has allowed a more nuanced view of location-sharing as a social practice that is complicated by its sociality: friend circles change over time, and negotiating the sociality of friendship via location-sharing presents as an emerging area of research in HCI.

It is striking that the practice of sharing location with ‘old’ friends was so common, in that there was no direct benefit from it. Yet, participants like James expressed that it would be ‘rude’ to delete them from the list, so that was why he kept them. Other participants simply said they might meet up with these people one day in the future, and that was the reason they kept them. It is worth noting that this kind of potential use aligns well with Bowker’s theory of the epoch of potential memory (Bowker 2008) which will receive more detailed coverage in the following subsections. Such broad self-disclosure contradicts much of the previous literature that highlighted numerous complex sharing limitations that people are willing to do, both in terms of people and granularity of location precision (Jedrzejczyk et al. 2010; Toch et al. 2010b; Sadeh et al. 2007; Tsai et al. 2009). While acknowledging that it would be ‘creepy’ to show up at a place where an acquaintance was located without communicating that first, some participants did have social structures in place that justified exactly that. And as such, privacy boundaries did not feel violated or even pushed.

5.2 Sharing Location to Build Trust

Location-sharing might seem like a manifestation of trust and solidification of a social connection. Yet it was apparent from the interviews that people often started sharing continuous location with others even as relationships were just being formed (i.e., in the absence of durably or time-tested trust). Most had started sharing with their roommates within the first weeks of sharing a home, and partners had started sharing within the first months of dating. In other words, trust was being built up *while* they were using location-sharing, not *before*. So the act of accepting a sharing request was not a display of trust, but in fact, an indication that they might trust them in the near future. Accordingly, location-sharing became a tool in social trust-building.

A now classic guideline for sociotechnical systems is consistent transparency, or as Erickson and Kellogg referred to, social translucence (Erickson and Kellogg 2000): making sure that the functionality is obvious, that the user is aware of technology behavior and activities, such as who they are sharing exactly what type of location with (i.e. granularity). And from a narrow design perspective location-sharing, as implemented in the two most common systems - *Find My* and *Life 360* - these are socially translucent: the user can at any point see who can see their location, and they can remove any unwanted sharers. Yet, on the ground, the tools clearly lacked consistent transparency because of the background nature of the apps and the actual practices that people would go days without looking at it, and sometimes forget that it even existed. Instead, the only consistent transparency

emerged through the mutual trust that they had with others, or was in the process of building up. We now continue relating, in more detail, our results to previous literature and particularly the notion of safety, which was the core reasoning to the participants' motivations.

5.3 Location-Sharing and the Construction of a “Sense” of Safety

The current location-sharing tools that we investigated, contradict significantly the predictions and prescriptions from the first decade of 2000, when location-sharing prototypes were first envisioned and developed (Iachello et al. 2005; Tsai et al. 2009; Consolvo et al. 2005; Benisch et al. 2011; Brush et al. 2010; Patil et al. 2015); even commercial applications would notify people regularly that they were sharing location with others (Holson 2007). Initially, many of these prototypes were developed for assisting practical activities, such as informing loved ones of how close they were to being home, and for meeting up (Tsai et al. 2009). Some early commercial products were marketed for keeping children safe, which emphasized a model of surveillance, rather than mutual sharing (Reardon 2009; Talor 2015).

However, as we pointed out in the results section, location-sharing among our participant was often motivated by the notion of safety; yet, none of the participants had any examples of situations when they had actually been in situations where their safety was on the line, and where friends could have ‘saved’ them, despite us asking participants to think back further than the two weeks of the study. Instead, the continuous location-sharing provided the people with a *sense* of personal safety and sometimes also a sense of security (defined as a broader level of protection from outside harm such as threats of violence, accidents and natural disasters (Coursen 2016). And even if it was clear that they were not necessarily safe from harm through location-sharing, it was a reasonable argument for them to use to provide friends and acquaintances with their continuous location.

Although safety was not core to how they used the location-sharing in most everyday activities, it was at the top of their mind for motivation, and as an “excuse” for having it in place, particularly with their local friends and roommates. It is therefore interesting how none of the above-cited studies mentioned safety as even a minor motivation for using such a sharing tool, contrasting our participants' motivations for sharing. As safety emerges as a prominent domain within HCI, we call for explicit, safety-related research about location-sharing technologies such as those we have analyzed in this paper.

Our participants' boundaries were miles beyond the predicted privacy limitations that were taken for granted in earlier research just 10 years ago. The unique situation of Generation Z probably plays an important role: This generation, and in particular those aged 22 and below, has a crucial age frame because they were the first generation whose first mobile phone, (often in middle school, around 10-12 years old) was a *smartphone* (McDonald et al. 2024). Only a handful of our participants had briefly had non-smartphones, often a hand-me-down from a family

member, for very explicit use at key times. Their experiences were solely with phones that had a wide range of functionalities, and they were quickly introduced to the world of location-based services and being available anywhere, anytime. However, we are of course aware that we cannot generalize about all urban college students as such, but from this small sample, it was evident that location-sharing was highly common among them and their peers and is here to stay. We note that our findings highlight the needed for sustained research engagement with young adults about the use of location-sharing technologies in their daily lives. Such a population emerges as an important population for HCI researchers and designers to engage with because it is unclear that the sociotechnical ecology in which prior findings about location-sharing were founded is sufficiently similar to the ecology in which today's young adults live their lives.

6 Implications

We continue by providing two implications of our research. These implications are directed toward fostering more solid, more robust theoretical foundations for understanding location-sharing practices in relation to the emergent sociotechnical ecology of the 2020s.

6.1 Acclimating to a New Safety Landscape

As described in Section 2, a great deal of research on location-sharing practices occurred alongside a sea of changes in the commonness of location-sharing services, devices, and platforms. Much of this work occurred in the early 2000s (i.e., around the times that Google Latitude and Apple's "Find My Friends" were released in 2005 and 2011, respectively). Yet the profound changes in the broader sociotechnical ecology (e.g., the saturation of smartphone use, the availability and even factory-inclusion of location-sharing technologies in smartphones, the widespread adoption of surveillance capitalistic data practices, etc.) highlight the need to reassess historical knowledge about location-sharing. Such a need is obvious from two perspectives. We describe these perspectives below.

6.1.1 Prior Work Contradictions

First, much of the prior work about location-sharing has focused on complicated routines or practices that users develop, or are expected to develop, to manage their privacy (e.g., Benisch et al. 2011; Iachello et al. 2005; Jedrzejczyk et al. 2010; Lederer et al. 2003; Sadeh et al. 2007; Toch et al. 2010a; Tsai et al. 2009). Based on our findings, such complexity may be a thing of the past. Just like with other technologies and data practices that have become more or less normalized over the past twenty years (e.g., Seberger et al. 2022b), contemporary college-aged people have grown up alongside location-sharing technologies. The worlds into which

they were “thrown” (Heidegger 1962) included location-sharing services.¹³ Such inclusion in young adults’ condition of thrownness differentiates young adults from previous generations. For example, members of older generations (e.g., elder Millennials, Gen X, Baby Boomers) have, at one time or another, had to re-calibrate their expectations of daily life alongside, or in reaction to, the development and release of novel technologies. The internet, mobile phones and smart phones representing examples of technologies whose functionality changed everyday practices significantly – up to and including the normalization of sharing evermore types of contextually-dependent data and information. Such normalization (e.g., (Seberger et al. 2022b)) may prevent people from thinking critically about the downstream effects of continuous sharing (cf (Seberger et al. 2021; Seberger et al. 2023)). After all, technological developments have rendered a world profoundly different in 2023, from the set of worlds that might have comprised the early 1980s (i.e., the lower boundary of the Millennial generation) or the late 1990s and early 2000s (i.e., the lower boundary of Gen Z).

Assuming the roughly homogenized perspective of such older generations – a homogenized perspective we manufacture here for purposes of illustration – such re-calibrations have obviously occurred in relation to the proliferation of location-sharing services. If one came of age during the largely *un*-networked 1990s, for example, the emergence of location-sharing technologies necessarily challenged assumptions about daily life (cf (Seberger and Bowker 2021)) that grew out of historical periods in which automated location-sharing was quite rare. From this perspective, it is possible that lived experiences of researchers who are Millennials, members of Generation X, or Baby Boomers present implicit confounds to understanding the lived worlds encountered by members of Generation Z or Generation Alpha. For members of older generations (e.g., Millennials, Generation X, etc.), our having borne witness to the emergence of technologies that are simply “natural” for Generation Z and Generation Alpha makes it especially difficult to understand just how natural such technologies are for these generations: what older generations witnessed as novel, younger generations simply *receive*.

We suggest that CSCW and HCI’s collective knowledge base about the use and perception of location-sharing services needs updating. This proposition is partly inspired by Serres, who in 2015 argued that Millennials live in functionally different worlds than do members of previous generations (e.g., Gen X, Baby Boomers, etc.). The vast differences between sociotechnical ecologies extends the argument to include the next generations (e.g. Gen Z, Gen Alpha, etc.). And as the sociotechnical ecology changes across time, so do the implicit expectations that people have about daily life in such an ecology. As such, we might state that time is medium of radical change (Castoriadis 1987). As the times (i.e., the sociotechnical

¹³ The Heideggerian notion of thrownness (i.e., having been *thrown*) refers to the fact that each of us is born into a particular historical era. While obviously not chosen by its recipient, the characteristics of such an era (e.g., the late 1990s, etc.) impact people’s expectations of daily life.

conditions of daily life) change or manifest in the form of technologically radical change, we must expect that the results of prior work require updating to account for such change.

6.2 Safety in the Epoch of Potential Memory

Our second implication relates to the theoretical framework of Epoch of Potential Memory. As an implication, we call for sustained ontological analysis of “safety” within a sociotechnical context. We observed that participants frequently shared their location with peers in order to foster a “sense” of safety. Such a sense of safety is necessarily future-oriented: it is *aspirational*. That is, participants described sharing their location information with peers in order to protect themselves against harm *should* harm befall them. From this we can infer that temporality may constitute an important facet of knowledge production about location-sharing practices. Such temporality indicates “safety” as a form of speculative vulnerability (Seberger et al. 2022a).

Bowker’s theorization of an “epoch of potential memory” (Bowker 2008) provides a useful framework for situating the temporality of participants’ “sense” of safety in relation to location-sharing; it also sheds light on the forms of vulnerability that may arise through “senses” of safety fostered through location-sharing apps. To paraphrase, Bowker (2008) defines the epoch of potential memory as a historical period in which data is collected for future use should the need for that data arise. In the epoch of potential memory, all is datafied precisely because the results of such datafication *may be useful* in the future. As such, location tracking and recording by service providers for potential future use, is also an example of this.

That participants ideate or contextualize their safety in the near future tense in which data about their location *may* be relevant speaks volumes about contemporary sociotechnical conditions. We noted, for example, that the majority of participants whose use of location-sharing apps gave them a sense of safety did not actually describe safety, *per se*, but instead the ability of others to locate them after they had already come to harm. Such aspirational safety stands in stark contrast to the image of “guardian angel[s]” described by one participant. These angels that location-sharing apps constitute are necessarily retrospective angels: like Benjamin’s angel of history (Benjamin 2020), such apps construct those with whom location is shared as possessed of a backwards gaze: post hoc witnesses to the present tense.¹⁴ Indeed, we noted at least one instance in which friends who received location updates from their peers were functionally unable to do anything about their peer’s safety when that peer left their phone in an Uber. Such post hoc witnesses are assumed to be

¹⁴ It is worth noting that such an understanding of temporality in relation to location-sharing apps aligns with Seberger’s theory of rhythmic diminution in relation to app culture and the archive of networked devices (Seberger 2022). Actions are taken in the experiential present tense in order to facilitate computational futures that occur at rhythms far too speedy to be experienced by people *qua* people.

benevolent. Yet through the lens of Bowker's epoch of potential memory (Bowker 2008), the benevolence of such witnesses is determined by context: the conditions in which location data are analyzed should the "need" for such analysis arise.

While participants discussed such a "sense" of safety, it may be more accurate to describe the underlying affective condition (i.e., that which allows for such a "sense" of safety) as one in which people enroll in digital ecologies and match their expectations to such digital ecologies. Novel characteristics of technologies (e.g., the *pervasiveness* of location-sharing apps) color and change the discourses that precede such technologies. The meaning and temporality of "safety" changes as a function of the infrastructures that mediate daily life. What we find in relation to college-aged participants' use of location-sharing apps is an aspirational sense of safety that exists solely in relation to the data-driven representation of the worlds in which young people live their lives.

The deployment of such humanities-oriented theoretical perspectives is essential if we are to demonstrate suitably systematic care for our younger generations. It is essential because in the absence of such broader considerations, it is likely that the forms of "safety" constructed through the proliferation of novel technologies are not the same as those we might construct through the analysis of data collected some twenty years ago. It may be a category error to claim that "safety" means the same thing for individuals or groups belonging to different generations, where such generations are differentiated primarily through comparison of the sociotechnical ecologies into which they were born.

6.3 Limitations

Interpretive caution should be exercised when making use of findings presented in this paper. Our study was a qualitative and exploratory study (n=15). Such a sample size is adequate to the task of exploratory consideration of location-sharing in social contexts. By virtue of its design and goals, findings presented are not strictly generalizable (or impacted by the burden of positivist generalizability). The phenomena we identified in this paper evidence a need for continued work in the area of location-sharing as it relates to privacy, trust, and safety: it identifies wrinkles in the contemporary understanding of location-sharing, but, again, we make no claims toward formal generalizability. Below, we describe some of the factors that should be taken into consideration when building upon or otherwise involving findings presented here in research.

While our participant sample seems very diverse, both in terms of cultural and socioeconomic background, all but one student were enrolled at private universities. Moreover, the study took place in a unique US context, New York City, where both student lifestyle and urban structure differ from most other areas in the world.

Our racial and ethnic diversity as such also suffered, the only African-American participant we recruited dropped out halfway through the study and was therefore not included. The ethnic diversity is therefore limited to White American, Asian and Asian-American, Hispanic, and first or second-generation Southern European-

American. As such, most of the students in the study were highly “privileged” students; yet it was clear that they had similar concerns and challenges as most other college students in the world in terms of academic performance, friendships, romantic relationships and sometimes bullying and mental health issues. But it means that we of course cannot generalize from the data, and we do not claim to be able to broaden our results to other socio-economic groups, students living in rural or suburban areas, or non-US context. They are rooted in New York students’ unique setting.

Finally, we recognize that parental monitoring was a crucial dimension brought up by some of the participants. With this acknowledgment, we believe that this is a pertinent issue better explored as a standalone phenomenon alongside peer location-sharing, especially because, in peer location-sharing, there are little to no added power asymmetries as there may be in a family structure. As such, we are exploring parental dynamics of location-sharing elsewhere.

6.4 Future Studies

While this study has provided insights into use of location sharing among a fairly narrow population, future work could for diversity focus on a broader set of people, such as young professionals, families and specific work settings, particularly due to a potentially different set of privacy preferences between these populations. Another level of insights could be gained through log-data analysis, through gaining device specific location sharing data, leading to more precise recording of device interaction. Previous research has looked at micro-interactions (Ferreira et al. 2014), and a similar tool could report exact measures of interaction, bypassing the need for participants to self-report. Finally, future work should look into privacy behavior in regards to people sharing within hierarchical relationships such as teenagers living at home, sharing with their parents, or people sharing location in work settings with their employer.

7 Conclusion

Through analysis of data collected in the three-part study described above, we found that college-aged users of location-sharing apps engage with such apps for several reasons. Most obviously (and practically), participants use location-sharing apps to orchestrate social actions and memberships of social groups. They found a wide set of practical uses, including checking if their roommates were away so they could bring a partner over, and casually meeting for breakfast before class without explicit coordination. More profoundly though, participants described use of location-sharing apps as a means of producing a “sense” of safety. Yet it would be irresponsible if we, as CSCW researchers and practitioners, fail to meaningfully interrogate just what “safety” means in an emergent and growing sociotechnical ecology. Safety was the notion that others had roles as “guardian angles” of them,

with an implicit understanding that the tool provided mutual safety. Yet, the sharing had no time or granularity limitations, only person-to-person rules, which occasionally were outdated. Our findings highlight the need to refresh or update prior findings about the use of and approach to location-sharing. Such a need arises as an effect of generational change and the ongoing transformation of former familiar discourses.

Participants' strong narrative around safety was surprising in contrast to most previous research, which focused on providing an intricate set of sharing rules, concerned with sharing with whom, to which detail, and for which time-frame, emphasizing a strong sense of privacy protection. Instead, participants were adamant about the sense of safety – their own and that of others – that location-sharing provided them, and therefore kept sharing continuously. We provided a simple description for understanding users' "sense of safety" as it pertains to both (i) their own sense of safety, (ii) their sense of others' safety and (iii) others' reassurance of their safety. Yet, participants were not naive: they occasionally admitted that they knew the sense of safety the apps provided was not necessarily *actual* safety from harm, but instead accountability for any perpetrator, and as such perhaps even a deterrent for a potential violator. The casual joking that "at least they know where to look for the body", illustrated this stark realism in the participants' minds. To end on a lighter, more poetic note, referring back to Taylor Swift's lyrics at the beginning, the casualness with which the participants approached location sharing was something someone could easily have forgotten to turn off.

Acknowledgements

The author thanks Lena Mamykina and her research group for hosting her and providing early feedback on the study. The direct costs were covered by the first author's internal faculty resources.

Author Contributions

LB conducted the interviews with assistance of JC. LB, JC and AM conducted the analysis, with the majority of work by LB. JS assisted in paper writing and theoretical foundations, but the majority of the paper was written and edited by LB. All authors reviewed the manuscript before submission.

Funding

Open access funding provided by IT University. This research was partly funded by The Danish Ministry of Higher Education and Science, which funded the first author's sabbatical at Columbia University in 2023, during which this study was conducted.

Data Availability

No datasets were generated or analysed during the current study.

Declarations

Competing Interests

The authors declare no competing interests.

Open Access

This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Ackerman, Mark S. 2000. The intellectual challenge of CSCW: The gap between social requirements and technical feasibility. *Human-Computer Interaction* 15 (2–3): 179–203. https://doi.org/10.1207/S15327051HCI1523_5.
- Bales, Elizabeth, Kevin A. Li, and William Griwsold. 2011. Couplevibe: Mobile implicit communication to improve awareness for (long-distance) couples. In *Proceedings of the ACM 2011 conference on Computer Supported Cooperative Work*, 65–74. <https://doi.org/10.1145/1958824.1958835>.
- Barkhuus, Louise. 2012. The mismeasurement of privacy: Using contextual integrity to reconsider privacy in HCI. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 367–376. <https://doi.org/10.1145/2207676.2207727>.
- Barkhuus, Louise, Marek Bell, Scott Sherwood, Malcolm Hall, and Matthew Chalmers. 2008. From awareness to repartee: Sharing location within social groups. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 497–506. <https://doi.org/10.1145/1357054.1357134>.
- Barkhuus, Louise, Matthew Chalmers, Paul Tennent, Malcolm Hall, Marek Bell, Scott Sherwood, and Barry Brown. 2005. Picking pockets on the lawn: The development of

- tactics and strategies in a mobile game. *International conference on ubiquitous computing* 358–374. https://doi.org/10.1007/11551201_21.
- Barkhuus, Louise, and Anind K. Dey. 2003. Location-based services for mobile telephony: A study of users' privacy concerns. *Proceedings of the INTERACT 2003, 9TH IFIP TC13 international conference on human-computer interaction* 3: 702–712.
- Barkhuus, Louise, and Paul Dourish. 2004. Everyday encounters with context-aware computing in a campus environment. *International conference on ubiquitous computing* 232–249. https://doi.org/10.1007/978-3-540-30119-6_14.
- Barkhuus, Louise, and Juliana Tashiro. 2010. Student socialization in the age of facebook. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 133–142. <https://doi.org/10.1145/1753326.1753347>.
- Beck, Julie. 2019. It's 10 p.m. do you know where your friends are? Constant location sharing is now the norm for some friend groups. <https://www.theatlantic.com/family/archive/2019/08/why-people-use-find-my-friends/596386/>. Accessed 23 March 2026.
- Bell, Marek, Matthew Chalmers, Louise Barkhuus, Malcolm Hall, Scott Sherwood, Paul Tennent, Duncan Rowland, Steve Benford, and Mauricio Capra. 2006. Interweaving mobile games with everyday life. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 417–426. <https://doi.org/10.1145/1124772.1124835>.
- Bellini, Rosanna, Emily Tseng, Nora McDonald, Rachel Greenstadt, Damon McCoy, Thomas Ristenpart, and Nicola Dell. 2021. “So-called privacy breeds evil”: Narrative justifications for intimate partner surveillance in online forums. In *Proceedings of ACM human computer interaction* 4(CSCW3). <https://doi.org/10.1145/3432909>.
- Benisch, Michael, Patrick Gage Kelley, Norman Sadeh, and Lorrie Faith Cranor. 2011. Capturing location-privacy preferences: Quantifying accuracy and user-burden tradeoffs. *Personal and Ubiquitous Computing* 15: 679–694. <https://doi.org/10.1007/s00779-010-0346-0>.
- Benjamin, Walter. 2020. *Theses on the philosophy of history*, 255–263. Routledge.
- Boesen, Julie, Jennifer A. Rode, and Clara Mancini. 2010. The domestic panopticon: Location tracking in families. In *Proceedings of the 12th ACM international conference on ubiquitous computing*, 65–74. <https://doi.org/10.1145/1864349.1864382>.
- Bowker, Geoffrey C. 2008. *Memory practices in the sciences*. MIT Press.
- Braun, Virginia, and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3 (2): 77–101.
- Brush, A. J. Bernheim, John Krumm, and James Scott. 2010. Exploring end user preferences for location obfuscation, location-based services, and the value of location. In *Proceedings of the 12th ACM international conference on ubiquitous computing*, 95–104. <https://doi.org/10.1145/1864349.1864381>.
- Castoriadis, Cornelius. 1987. *The imaginary institution of society*. MIT Press.
- Chatterjee, Rahul, Periwinkle Doerfler, Hadas Orgad, Sam Havron, Jackeline Palmer, Diana Freed, Karen Levy, Nicola Dell, Damon McCoy, and Thomas Ristenpart. 2018. The spyware used in intimate partner violence. *IEEE Symposium on Security and Privacy (SP)* 441–458. <https://doi.org/10.1109/SP.2018.00061>.

- Christensen, Tamlin Conner, Lisa Feldman Barrett, Eliza Bliss-Moreau, Kirsten Lebo, and Cynthia Kaschub. 2003. A practical guide to experience-sampling procedures. *Journal of Happiness Studies* 4 (1): 53–78.
- Consolvo, Sunny, Ian E. Smith, Tara Matthews, Anthony LaMarca, Jason Tabert, and Pauline Powledge. 2005. Location disclosure to social relations: Why, when, & what people want to share. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 81–90. <https://doi.org/10.1145/1054972.1054985>.
- Coursen, Spencer. 2016. Safety vs. security: Understanding the difference may soon save lives. <https://spencercoursen.medium.com/safety-vs-security-understanding-the-difference-may-soon-save-lives-71ac2e7517c3>. Accessed 23 Mar 2026.
- Davis, Isabella S., Makayla T. Thornburg, Herry Patel, and William E. Pelham III. 2023. Digital location tracking of children and adolescents: A review and theoretical model. *OSF Preprints*. <https://doi.org/10.31219/osf.io/2undy>.
- Dungey, Claire Elisabeth. 2024. A bird's eye view: Care, control, and the use of surveillance apps among family members in Germany. *Sociología* 56 (4): 325–345.
- Edwards, Paul N. 2019. Infrastructuration: On habits, norms and routines as elements of infrastructure. *Thinking Infrastructures* 62:355–366.
- Erickson, Thomas, and Wendy A. Kellogg. 2000. Social translucence: An approach to designing systems that support social processes. *ACM Transactions on Computer-Human Interaction (TOCHI)* 7 (1): 59–83. <https://doi.org/10.1145/344949.345004>.
- Federal Communications Commission. 1999. Enhanced 911 - wireless services. <https://www.fcc.gov/general/enhanced-9-1-1-wireless-services>. Accessed 23 Mar 2026.
- Ferreira, Denzil, Jorge Goncalves, Vassilis Kostakos, Louise Barkhuus, and Anind K. Dey. 2014. Contextual experience sampling of mobile application micro-usage. In *Proceedings of the 16th international conference on human-computer interaction with mobile devices & services*, 91–100. <https://doi.org/10.1145/2628363.2628367>.
- Fildes, Jonathan. 2026. Mobiles navigate the future. <http://news.bbc.co.uk/2/hi/science/nature/6294287.stm>. Accessed 21 Jan 2026.
- Franz, Madelyn, and Andrew Laningham. 2022. Are location sharing features more than a convenient tool? <https://theharrispoll.com/briefs/location-sharing-features/>. Accessed 23 Mar 2026.
- Fraser, Cynthia, Erica Olsen, Kaofeng Lee, Cindy Southworth, and Sarah Tucker. 2010. The new age of stalking: Technological implications for stalking. *Juvenile and Family Court Journal* 61 (4): 39–55.
- Freed, Diana, Jackeline Palmer, Diana Minchala, Karen Levy, Thomas Ristenpart, and Nicola Dell. 2018. “A stalker’s paradise” how intimate partner abusers exploit technology. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 1–13. <https://doi.org/10.1145/3173574.31742>.
- Guha, Shion, and Stephen B. Wicker. 2015. Do birds of a feather watch each other? Homophily and social surveillance in location based social networks. In *Proceedings of the 18th ACM conference on computer supported cooperative work & social computing*, 1010–1020. <https://doi.org/10.1145/2675133.2675179>.

- Harrison, Steve, and Paul Dourish. 1996. Re-place-ing space: The roles of place and space in collaborative systems. In *Proceedings of the 1996 ACM conference on computer supported cooperative work*, 67–76. <https://doi.org/10.1145/240080.240193>.
- Harrison, Steve, and Deborah Tatar. 2008. Places: People, events, loci - the relation of semantic frames in the construction of place. *Computer Supported Cooperative Work* 17 (2–3): 97–133. <https://doi.org/10.1007/s10606-007-9073-0>.
- Heidegger, Martin. 1962. *Heidegger, being and time*. New York: Harper & Row.
- Holson, Laura M. 2007. Privacy lost: These phones can find you. <http://www.nytimes.com/2007/10/23/technology/23mobile.html>. Accessed 23 Mar 2026.
- Iachello, Giovanni, Ian Smith, Sunny Consolvo, Mike Chen, and Gregory D. Abowd. 2005. Developing privacy guidelines for social location disclosure applications and services. In *Proceedings of the 2005 symposium on usable privacy and security*, 65–76. <https://doi.org/10.1145/1073001.1073008>.
- Jedrzejczyk, Lukasz, Blaine A. Price, Arosha K. Bandara, and Bashar Nuseibeh. 2010. On the impact of real-time feedback on users' behaviour in mobile location-sharing applications. In *Proceedings of the sixth symposium on usable privacy and security*, 1–12. <https://doi.org/10.1145/1837110.1837129>.
- Jennings, Rebecca. 2023. Should we know where our friends are at all times? "I love you, now let me watch your location 24/7". <https://www.vox.com/culture/23742552/location-sharing-iphone-friends-privacy-risks>. Accessed 23 Mar 2026.
- Jones, Quentin, Sukeshini A. Grandhi, Loren Terveen, and Steve Whittaker. 2004. People-to-people-to-geographical-places: The p3 framework for location-based community systems. *Computer Supported Cooperative Work (CSCW)* 13: 249–282. <https://doi.org/10.1007/s10606-004-2803-7>.
- Klein, Heinz K., and Michael D. Myers. 1999. A set of principles for conducting and evaluating interpretive field studies in information systems. *MIS Quarterly* 67–93. <https://doi.org/10.2307/249410>.
- Knijnenburg, Bart P., Alfred Kobsa, and Hongxia Jin. 2013. Preference-based location sharing: Are more privacy options really better? In *Proceedings of the SIGCHI conference on human factors in computing systems*, 2667–2676. <https://doi.org/10.1145/2470654.2481369>.
- Lederer, Scott, Jennifer Mankoff, and Anind K. Dey. 2003. Who wants to know what when? Privacy preference determinants in ubiquitous computing. In *Extended abstracts of SIGCHI conference on human factors in computing systems*, 724–725. <https://doi.org/10.1145/765891.765952>.
- Li, Nan, and Guanling Chen. 2010. Sharing location in online social networks. *IEEE Network* 24 (5): 20–25. <https://doi.org/10.1109/MNET.2010.5578914>.
- Lyon, David. 2007. *Surveillance studies. An overview*. Polity.
- Mancini, Clara, Yvonne Rogers, Keerthi Thomas, Adam N. Joinson, Blaine A. Price, Arosha K. Bandara, Lukasz Jedrzejczyk, and Bashar Nuseibeh. 2011. In the best families: Tracking and relationships. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 2419–2428. <https://doi.org/10.1145/1978942.1979296>.
- Mcdonald, Nora, John S. Seberger, and Afsaneh Razi. 2024. For me or not for me? The ease with which teens navigate accurate and inaccurate personalized social media content. In

- Proceedings of the SIGCHI conference on human factors in computing systems*, 1–7. <https://doi.org/10.1145/3613904.3642297>.
- Mols, Anouk, Jorge Pereira Campos, and Jason Pridmore. 2023. Family surveillance: Understanding parental monitoring, reciprocal practices, and digital resilience. *Surveillance & Society* 21(4): 469–484. <https://doi.org/10.24908/ss.v21i4.15645>.
- Myers, Michael D., and Michael Newman. 2007. The qualitative interview in is research: Examining the craft. *Information and Organization* 17 (1): 2–26. <https://doi.org/10.1016/j.infoandorg.2006.11.001>.
- NY State, Division of Criminal Justice Services. 2026. New york state index crime. https://www.criminaljustice.ny.gov/tableau_index_crime.htm. Accessed 23 Mar 2026.
- Page, Xinru, Bart P. Knijnenburg, and Alfred Kobsa. 2013. What a tangled web we weave: Lying backfires in location-sharing social media. In *Proceedings of the 2013 conference on computer supported cooperative work*, 273–284. <https://doi.org/10.1145/2441776.2441808>.
- Page, Xinru, and Alfred Kobsa. 2009. The circles of latitude: Adoption and usage of location tracking in online social networking. *2009 International Conference on Computational Science and Engineering* 4:1027–1030. <https://doi.org/10.1109/CSE.2009.195>.
- Page, Xinru, and Alfred Kobsa. 2010. Navigating the social terrain with google latitude. *2010 iConference* 174.
- Patil, Sameer, Roberto Hoyle, Roman Schlegel, Apu Kapadia, and Adam J. Lee. 2015. Interrupt now or inform later? Comparing immediate and delayed privacy feedback. In *Proceedings of the SIGCHI ACM conference on human factors in computing systems*, 1415–1418. <https://doi.org/10.1145/2702123.2702165>.
- Pier, Ken. 1991. Locator technology in distributed systems: The active badge. In *Proceedings of the conference on organizational computing systems*, 285–287. <https://doi.org/10.1145/122831.122858>.
- Reardon, Marguerite. 2009. AT&T launches family-tracking service. <https://www.cnet.com/tech/mobile/at-t-launches-family-tracking-service/>.
- Rost, Mattias, Louise Barkhuus, and Henriette Cramer. 2013. Representation and communication: Challenges in interpreting large social media datasets. In *Proceedings of the 2013 conference on computer supported cooperative work*, 357–362. <https://doi.org/10.1145/2441776.2441817>.
- Rubin, Herbert J., and Irene S. Rubin. 2011. *Qualitative interviewing: The art of hearing data*. Sage.
- Sadeh, Norman, Jason Hong, Lorrie Cranor, Ian Fette, Patrick Kelley, Madhu Prabaker, and Jinghai Rao. 2007. Understanding and capturing people’s privacy policies in a people finder application. In *Proceedings of the ubicomp workshop on privacy*. <https://doi.org/10.1184/R1/6626354>.
- Schildt, Emily, Martin Leinfors, and Louise Barkhuus. 2016. Communication, coordination and awareness around continuous location sharing. In *Proceedings of the 2016 ACM international conference on supporting group work*, 257–265. <https://doi.org/10.1145/2957276.2957289>.

- Seberger, John S. 2022. Into the archive of ubiquitous computing: The data perfect tense and the historicization of the present. *Journal of Documentation* 78 (1): 18–37. <https://doi.org/10.1108/JD-11-2020-0195>.
- Seberger, John S., and Geoffrey C. Bowker. 2021. Humanistic infrastructure studies: Hyper-functionality and the experience of the absurd. *Information, Communication & Society* 24 (12): 1712–1727. <https://doi.org/10.1080/1369118X.2020.1726985>.
- Seberger, John S., Hyesun Choung, and Prabu David. 2023. Problematizing “empowerment” in HCAI. In *IFIP Conference on human-computer interaction*, 270–279. https://doi.org/10.1007/978-3-031-42286-7_15.
- Seberger, John S., Marissel Llavore, Nicholas Nye Wyant, Irina Shklovski, and Sameer Patil. 2021. Empowering resignation: There’s an app for that. In *Proceedings of the 2021 SIGCHI conference on human factors in computing systems*, 1–18. <https://doi.org/10.1145/3411764.3445293>.
- Seberger, John S., Ike Obi, Mariem Loukil, William Liao, David J. Wild, and Sameer Patil. 2022. Speculative vulnerability: Uncovering the temporalities of vulnerability in people’s experiences of the pandemic. In *Proceedings of the ACM on human-computer interaction* 6 (CSCW2): 1–27. <https://doi.org/10.1145/3555586>.
- Seberger, John S., Irina Shklovski, Emily Swiatek, and Sameer Patil. 2022. Still creepy after all these years: The normalization of affective discomfort in app use. In *Proceedings of the 2022 SIGCHI conference on human factors in computing systems*, 1–19. <https://doi.org/10.1145/3491102.3502112>.
- Seeburger, Jan, and Ronald Schroeter. 2009. Disposable maps: Ad hoc location sharing. In *Proceedings of the 21st annual conference of the Australian computer-human interaction special interest group: Design: Open 24/7*, 377–380. <https://doi.org/10.1145/1738826.1738902>.
- Smith, Ian, Sunny Consolvo, Anthony Lamarca, Jeffrey Hightower, James Scott, Timothy Sohn, Jeff Hughes, Giovanni Iachello, and Gregory D. Abowd. 2005. Social disclosure of place: From location technology to communication practices. *PERVASIVE’05* 134–151. https://doi.org/10.1007/11428572_9.
- Sukk, Marit, and Andra Siibak. 2021. Caring dataveillance and the construction of “good parenting”: Estonian parents’ and pre-teens’ reflections on the use of tracking technologies. *Communications* 46 (3): 446–467. <https://doi.org/10.1515/commun-2021-0045>.
- Talor, Dave. 2015. An overview of AT&T family map, 2015. <https://youtu.be/HdhOBZ66mq4si=9ZVU4JImvKJ0Pr4v>. Accessed 23 Mar 2026.
- Toch, Eran, Justin Cranshaw, Paul Hanks Drielsma, Janice Y Tsai, Patrick Gage Kelley, James Springfield, Lorrie Cranor, Jason Hong, and Norman Sadeh. 2010. Empirical models of privacy in location sharing. In *Proceedings of the 12th ACM international conference on ubiquitous computing*, 129–138. <https://doi.org/10.1145/1864349.1864364>.
- Toch, Eran, Justin Cranshaw, Paul Hanks-Drielsma, Jay Springfield, Patrick Gage Kelley, Lorrie Cranor, Jason Hong, and Norman Sadeh. 2010. Locaccino: A privacy-centric location sharing application. In *Proceedings of the 12th ACM international conference adjunct papers on ubiquitous computing-adjunct*, 381–382. <https://doi.org/10.1145/1864431.1864446>.

- Toch, Eran, Ramprasad Ravichandran, Lorrie Faith Cranor, Paul Hanks Drielsma, Jason I. Hong, Patrick Gage Kelley, Norman M. Sadeh, and Janice Y. Tsai. 2009. Analyzing use of privacy policy attributes in a location sharing application. In *SOUPS '09: Proceedings of the 5th symposium on usable privacy and security*. <https://doi.org/10.1145/1572532.1572571>.
- Tsai, Janice Y., Patrick Kelley, Paul Drielsma, Lorrie Faith Cranor, Jason Hong, and Norman Sadeh. 2009. Who's viewed you? The impact of feedback in a mobile location-sharing application. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 2003–2012. <https://doi.org/10.1145/1518701.1519005>.
- Wan, Lin, Claudia Müller, Volker Wulf, and David William Randall. 2014. Addressing the subtleties in dementia care: Pre-study & evaluation of a GPS monitoring system. In *Proceedings of the SIGCHI conference on human factors in computing systems*, 3987–3996. <https://doi.org/10.1145/2556288.2557307>.
- Widmer, Sarah, and Anders Albrechtslund. 2021. The ambiguities of surveillance as care and control: Struggles in the domestication of location-tracking applications by danish parents. *Nordicom Review* 42 (s4): 79–93. <https://doi.org/10.2478/nor-2021-0042>.
- Xie, Jierui, Bart Piet Knijnenburg, and Hongxia Jin. 2014. Location sharing privacy preference: Analysis and personalized recommendation. In *Proceedings of the 19th international conference on intelligent user interfaces*, 189–198. <https://doi.org/10.1145/2557500.2557504>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.